

Janushan Sivakrishna

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EDUCATION

Toronto Metropolitan University

Sept. 2023 – Apr. 2028

Bachelor's of Engineering – Mechanical Engineering

Toronto, ON

- Hands-on experience in Solid Mechanics Labs, Materials Science Labs, Control Systems Labs
- Relevant Courses: Solid Mechanics I, Mechanical Design I, Applied Thermodynamics, Control Systems

TECHNICAL SKILLS

3-D CAD Tools: SolidWorks, AutoCAD, Fusion 360

Programming & Analysis: MATLAB, Python, Ansys FEA, C/C++

Manufacturing Knowledge: Engineering Drawings, CNC Machining, 3-D Printing

Other Skills: Microsoft 365 (Excel, Word), Problem Solving, Technical Communication, Leadership, Teamwork

EXPERIENCE

Mechanical Design Intern

Mar. 2025 – May 2025

CPM Tools Industries Inc.

Toronto, ON

- Optimized three self-adjusting and ratchet mechanisms, improving tool efficiency by 10% through refined mechanical design.
- Produced detailed technical drawings and comprehensive BOM in SolidWorks, ensuring full manufacturing accuracy and material traceability.
- Conducted competitive analysis of four market competitors and developed a 5-page market expansion report, identifying three new product opportunities.

Competitions Coordinator

May 2024 – Present

Metropolitan Engineering Competition

Toronto, ON

- Planned and organized a hydraulic lift bridge design competition for over 80 undergraduate students, overseeing logistics, materials, and event flow.
- Created precise competition abstracts, rulebook, and assessment rubrics to ensure clear objectives and fair scoring aligned with OEC guidelines.
- Delivered the competition presentation to an audience of judges, industry professionals, and fellow competitors, effectively conveying design goals and event outcomes.

Automotive Service Technician

Jul. 2022 – Dec. 2022

Pennzoil

Toronto, ON

- Performed over 50 oil changes and inspections, using Droptop software for documentation and tracking.
- Assisted in tire changes and lubrication services to ensure safety and reliability.
- Organized and streamlined inventory of service products and tools, improving shop floor management.

PROJECTS

RC Rover

Apr. 2026 – May 2026

- Designed chassis, 3DOF robotic arm, and claw mechanism in Onshape and SolidWorks, then 3D printed components using Bambu Lab printers.
- Integrated ESP32, servo motors, and L298N motor driver for vehicle mobility and arm actuation.
- Programmed in C++ (Arduino IDE) with wireless DS4 controller for real-time navigation and claw operation.

BiathlONE Autonomous Robot

Jan. 2026 – Jan. 2026

- Designed and built an autonomous Arduino-based robot integrating ultrasonic, IR, and color sensors for real-time navigation, obstacle avoidance, and path selection.
- Developed embedded C++ code to implement state-based logic, sensor feedback control, and conditional decision-making for autonomous task sequencing.
- Optimized mechanical stability and sensor calibration through iterative testing to achieve consistent performance under competition time constraints.

Rankine Cycle Plant Design

Oct. 2025 – Dec. 2025

- Designed an 80 MWe / 25 MWth natural gas Rankine cycle power plant providing electricity and district heating for remote Canadian communities.
- Optimized thermal efficiency through 3 design iterations with a 4-stage turbine, reheat, and regenerative feedwater heating to reduce fuel consumption.
- Validated thermodynamic performance and fuel cost feasibility using MATLAB simulations integrated with the XSteam library.